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RUHR-UNIVERSITÄT BOCHUM

GLOBAL CHANGE HYDROLOGY (GCH)

Overview of WaterGAP3 model

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Chair of Engineering Hydrology and Water Resources Management

In General

- **Large-scale hydrological models are important tools to assess global water resources (water quantity & water quality)**
- **Two types of developments: Land Surface Models (LSF) and Global Hydrological Models (GHM)**
- **Further improvements by application of EO products and computational capacity**
- **„Grand challenge“ (Wood et al. 2011): global models with spatial resolution of 1 km**

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Introduction

Area of application:

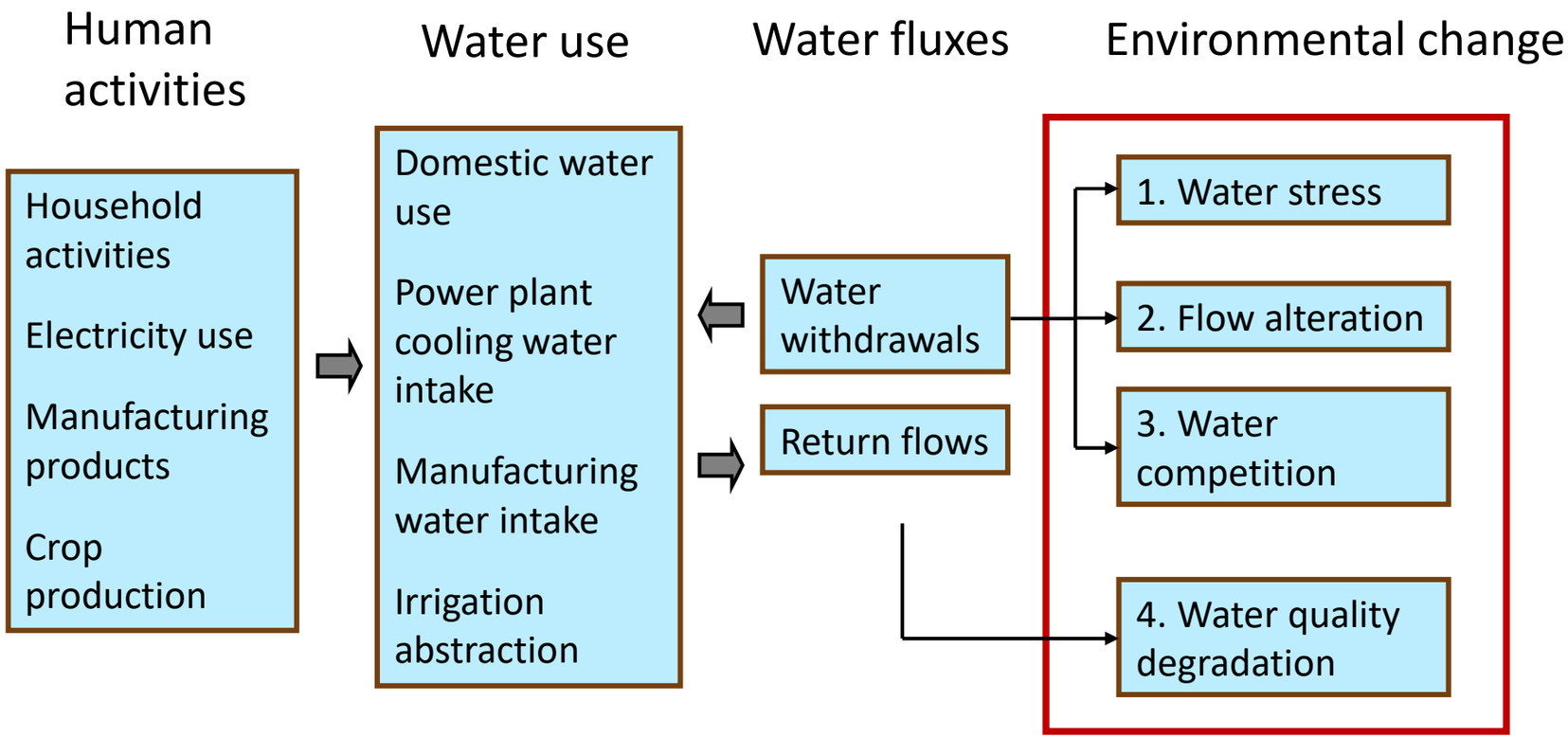
- Extremes (floods & droughts)
- Anthropogenic stressors / modifications
- Water scarcity
- Water quality degradation
- Other impact assessments

Sources of uncertainty:

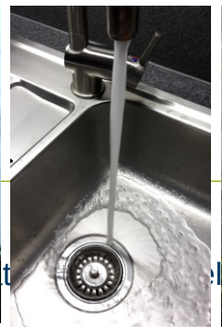
- Simple model structure
- Parameterization/conceptual parameters
- Limitations in calibration
- Missing field experience

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Current Research





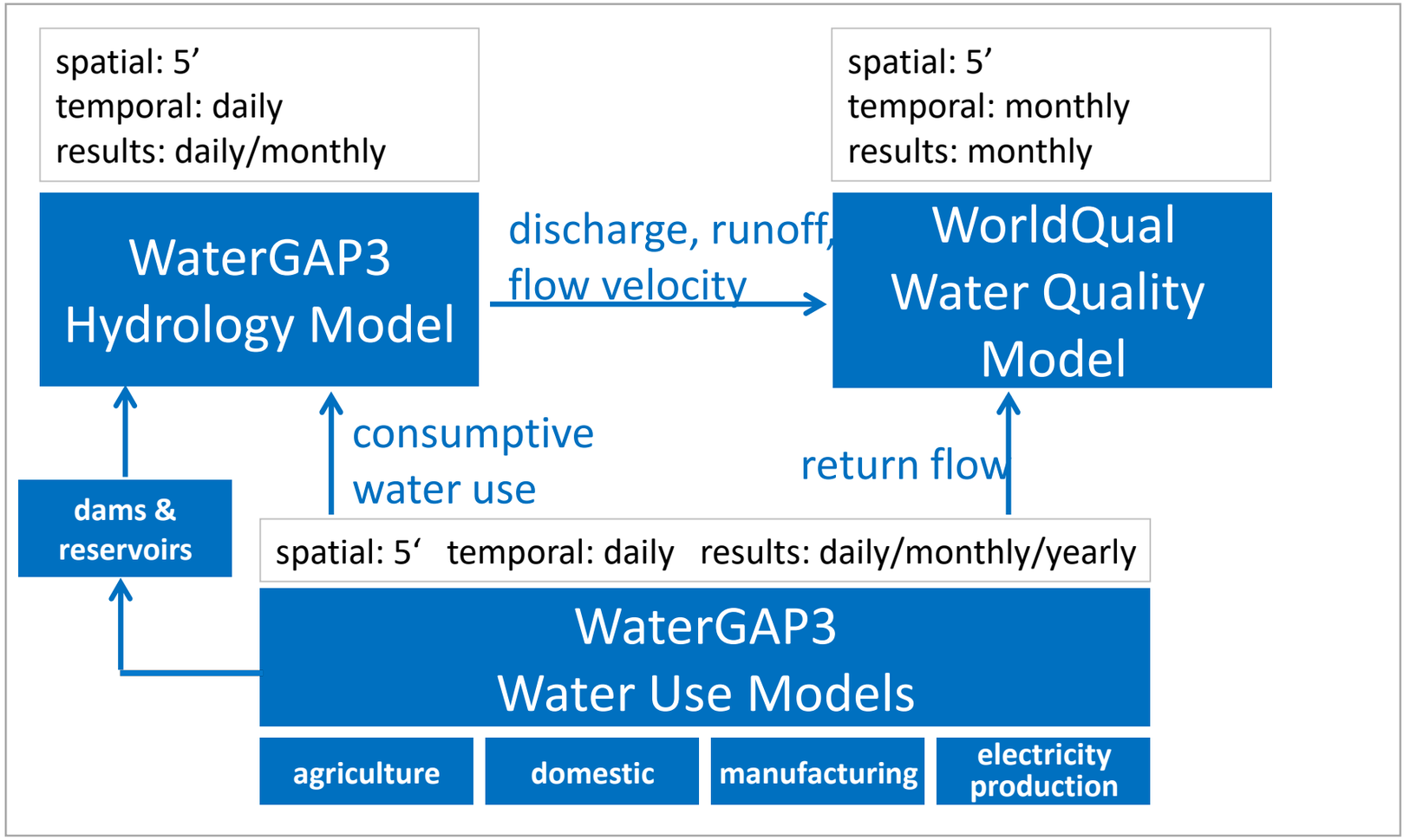






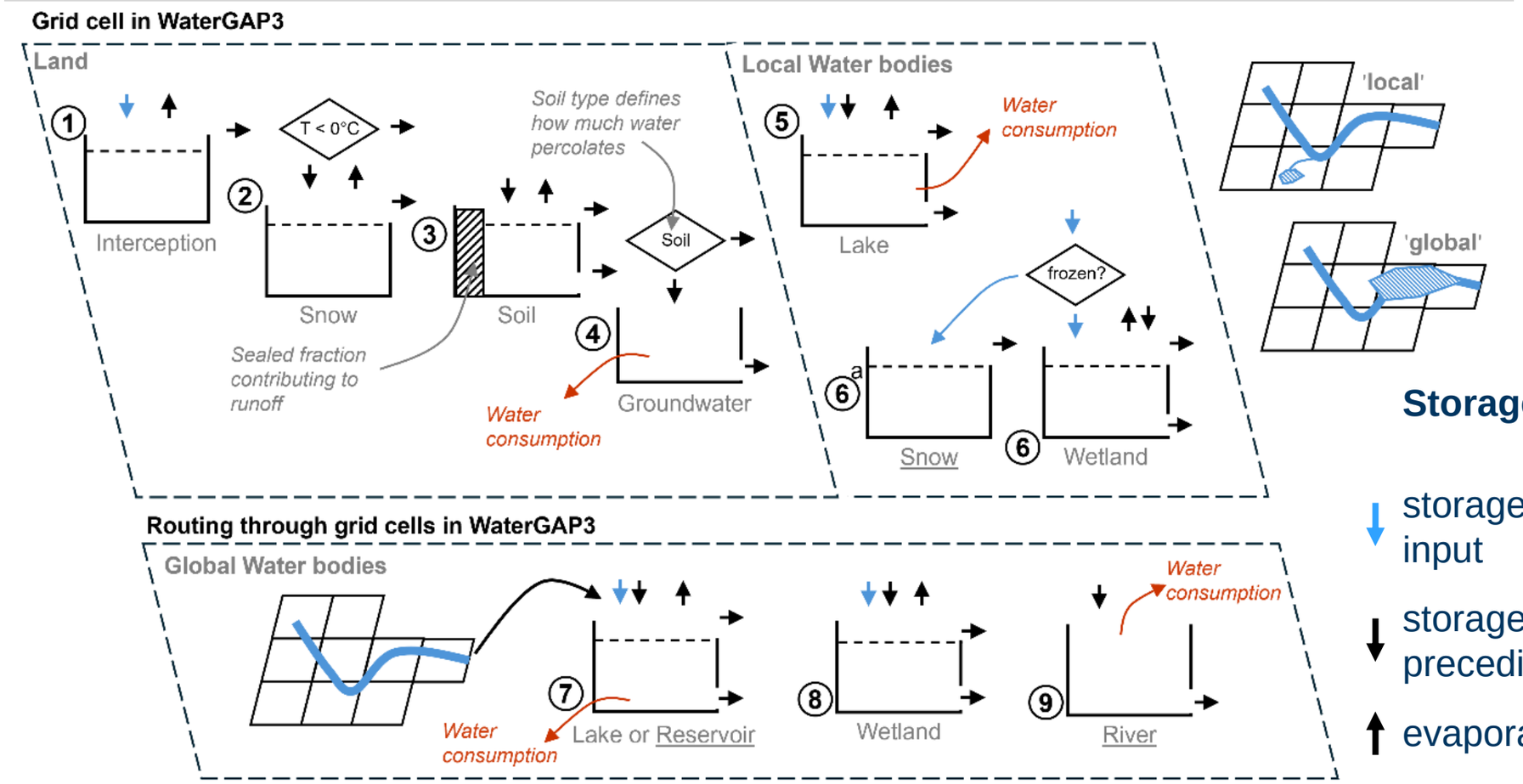
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WaterGAP3 Modelling Framework



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WaterGAP3: A Water Balance Model



Storages in WaterGAP3:

- storages that receive precipitation as input
- storages receive water from their preceding storage
- evaporation or sublimation processes

Kupzig & Flörke (2025)

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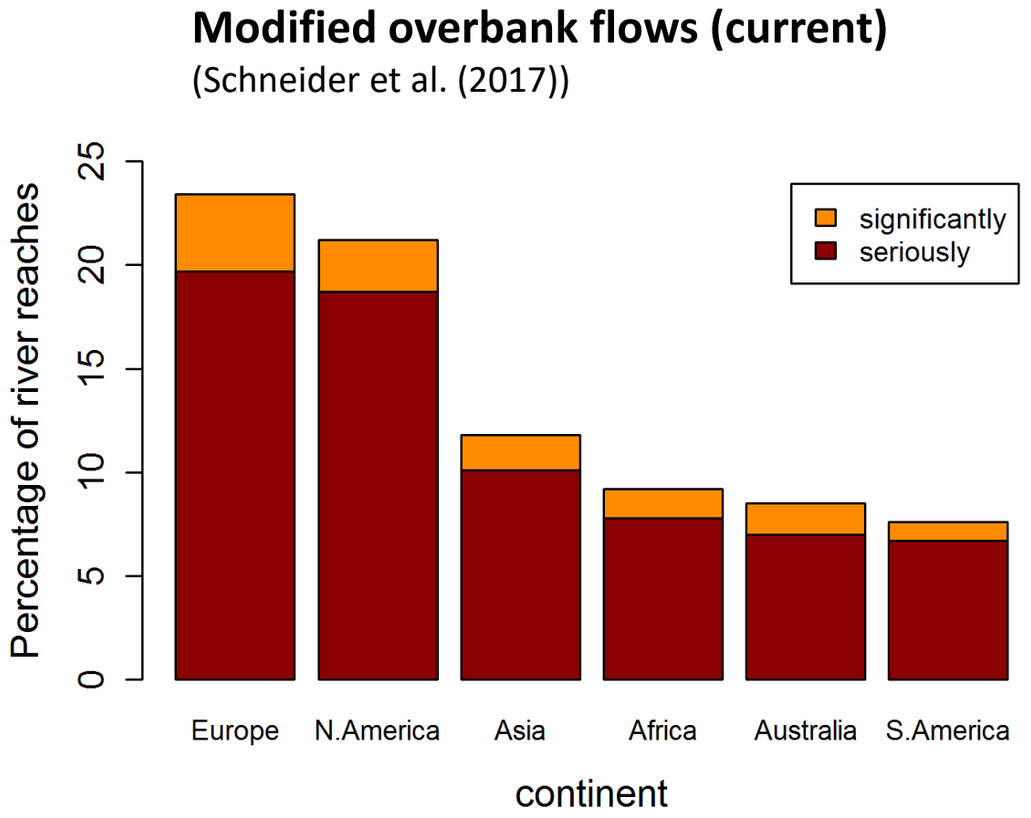
Anthropogenic Stressors



- Storage capacity of large dams: ~8300 km³
- Global water withdrawals : ~4000 km³/a
- Urban water transfer: 244 km³/a (480 cities)
- Point source wastewater intake: >450 km³/a (domestic, industry)

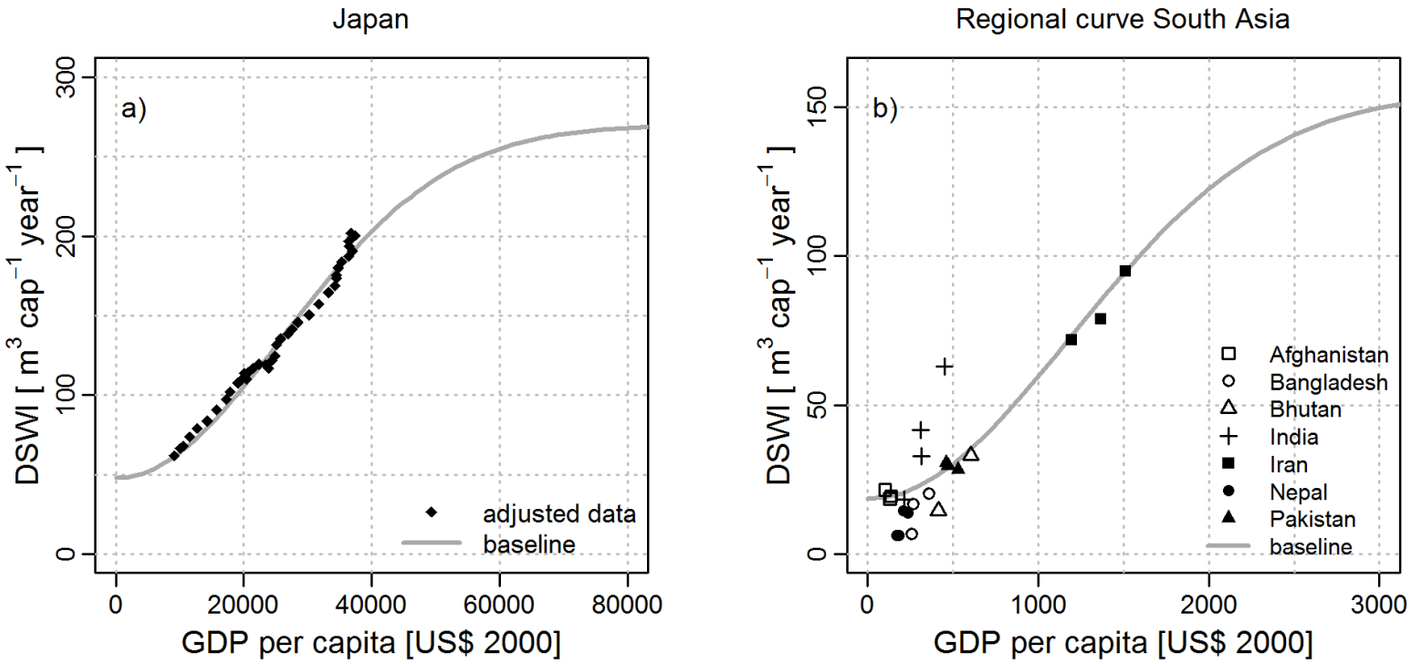
Reservoirs and Dams (GranD)

- Part of the hydrological model
- 6025 large dams, storage capacity ~6200 km³
- Reservoir management: Hanasaki approach for irrigation and non-irrigation water supply



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Domestic Water Use Model



$$DSWI = DSWI_{\min} + \alpha \cdot \left(1 - e^{-\gamma \cdot GDP^2}\right) \qquad \left[\frac{m^3}{cap \cdot year}\right]$$

a) Water withdrawals:

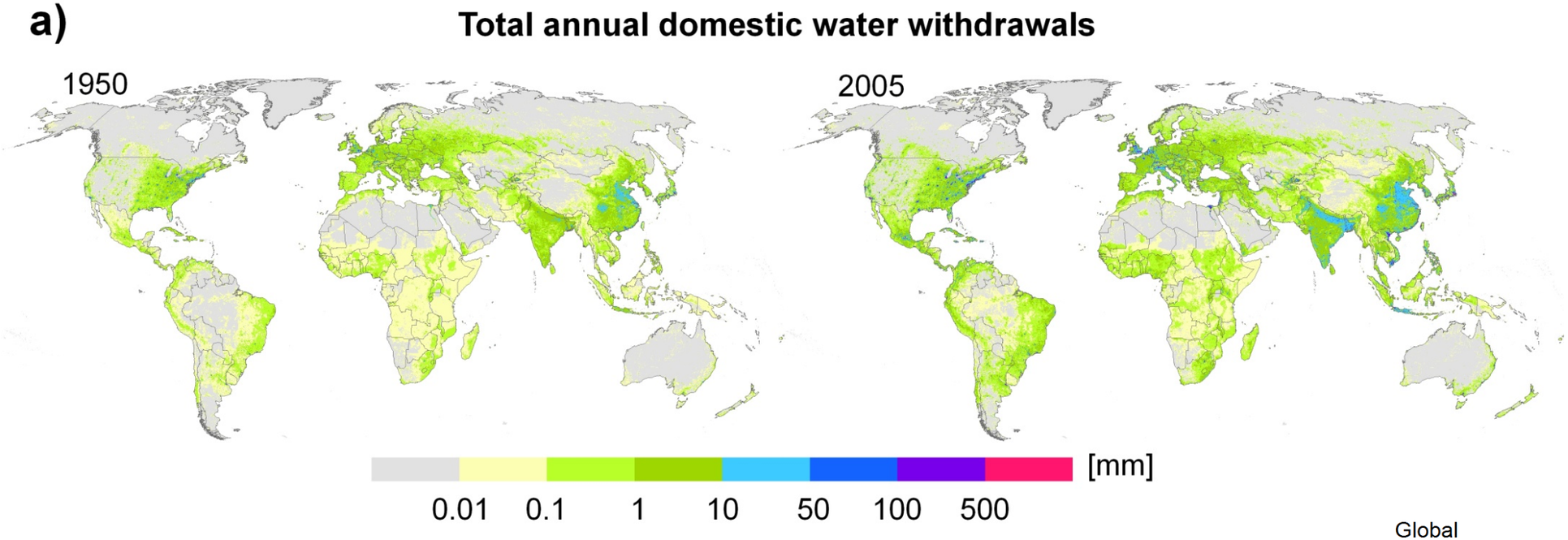
$$DWW = DWSI \cdot Tch_D \cdot population$$

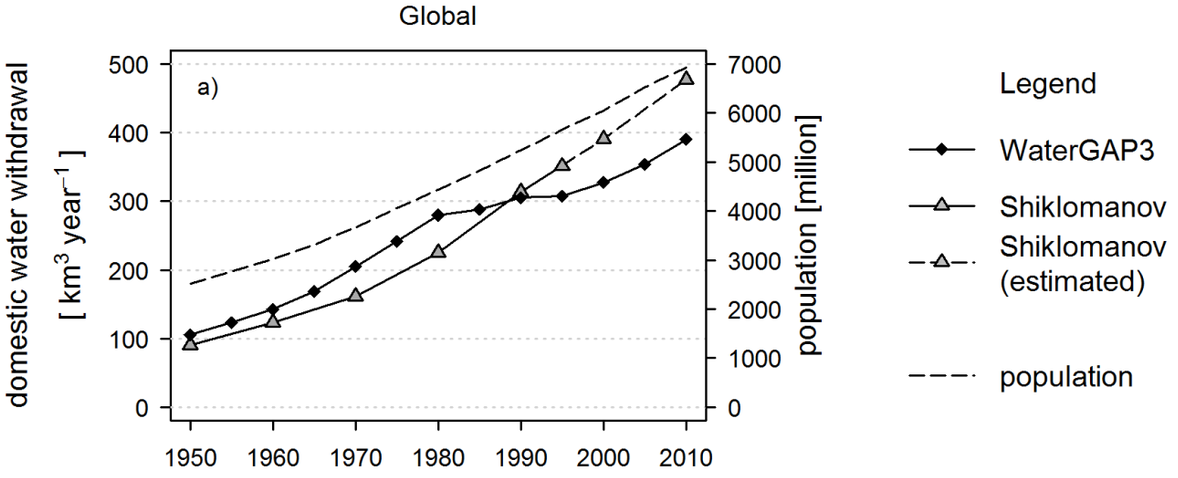
b) Water consumption:

- consumptive water-use coefficients (1950-1999)
- difference between water withdrawal and return flow (2000-2010)

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Domestic Water Withdrawals





Flörke et al. (2013)

Thermoelectric Water Use Model

a) Water withdrawals:

$$TPWW = \sum_{i=1}^{n} EP_i \cdot WWI_i(CS_i, PT_i) \cdot Tch_{TP}$$

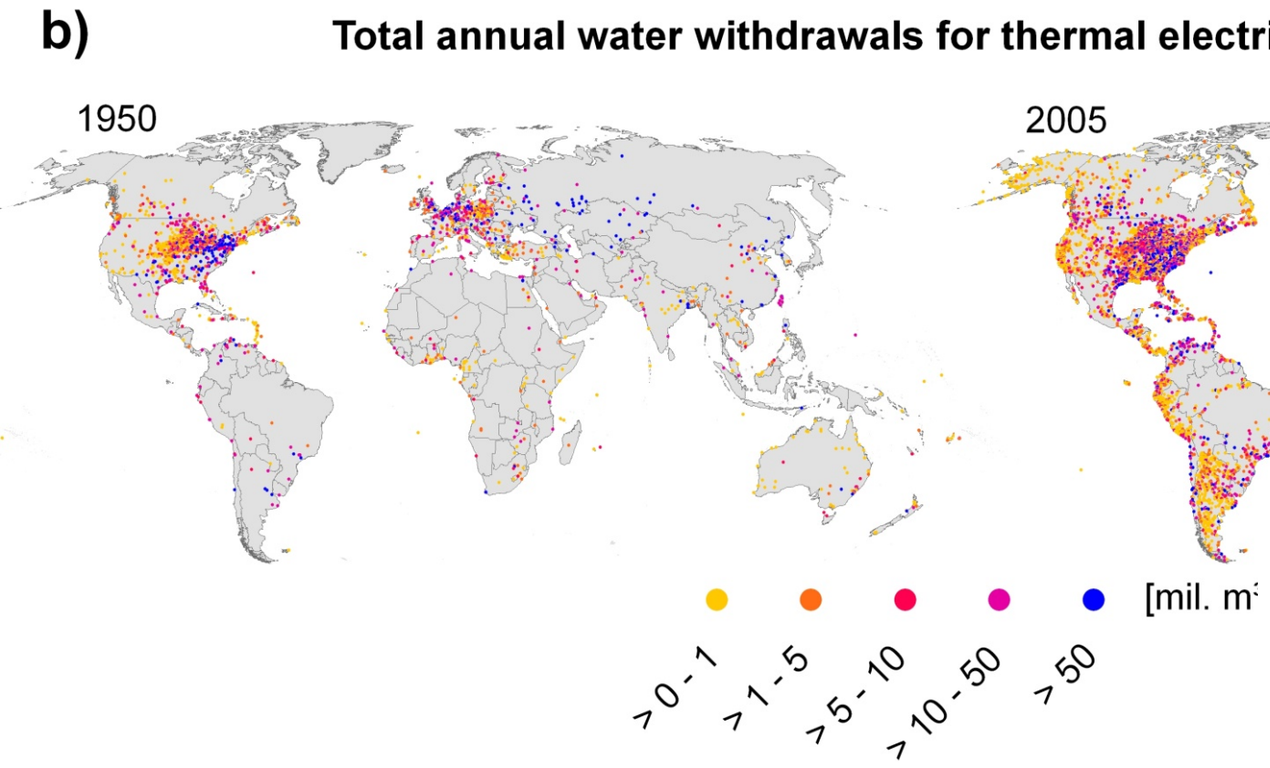
b) Water consumption:

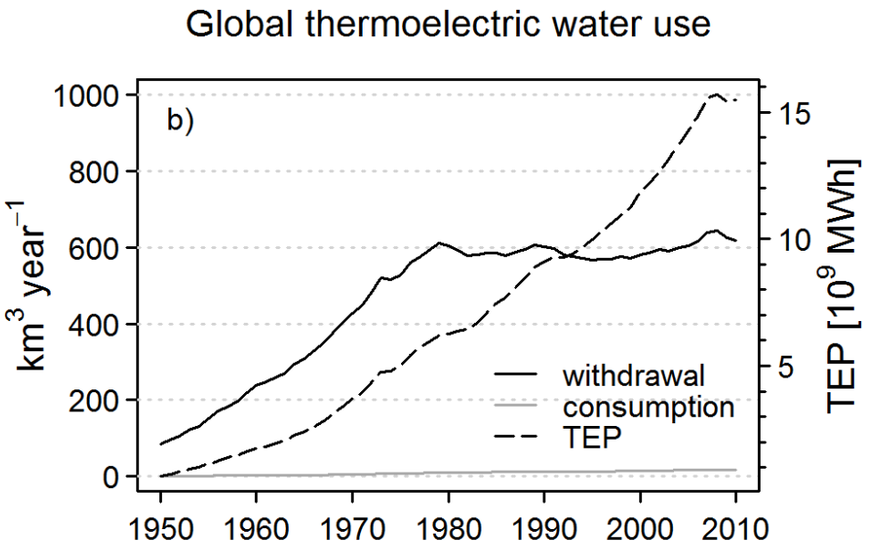
$$TPWC = \sum_{i=1}^{n} EP_i \cdot WCI_i(CS_i, PT_i) \cdot Tch_{TP}$$

14 combinations of plant type (PT) and cooling system (CS) are considered

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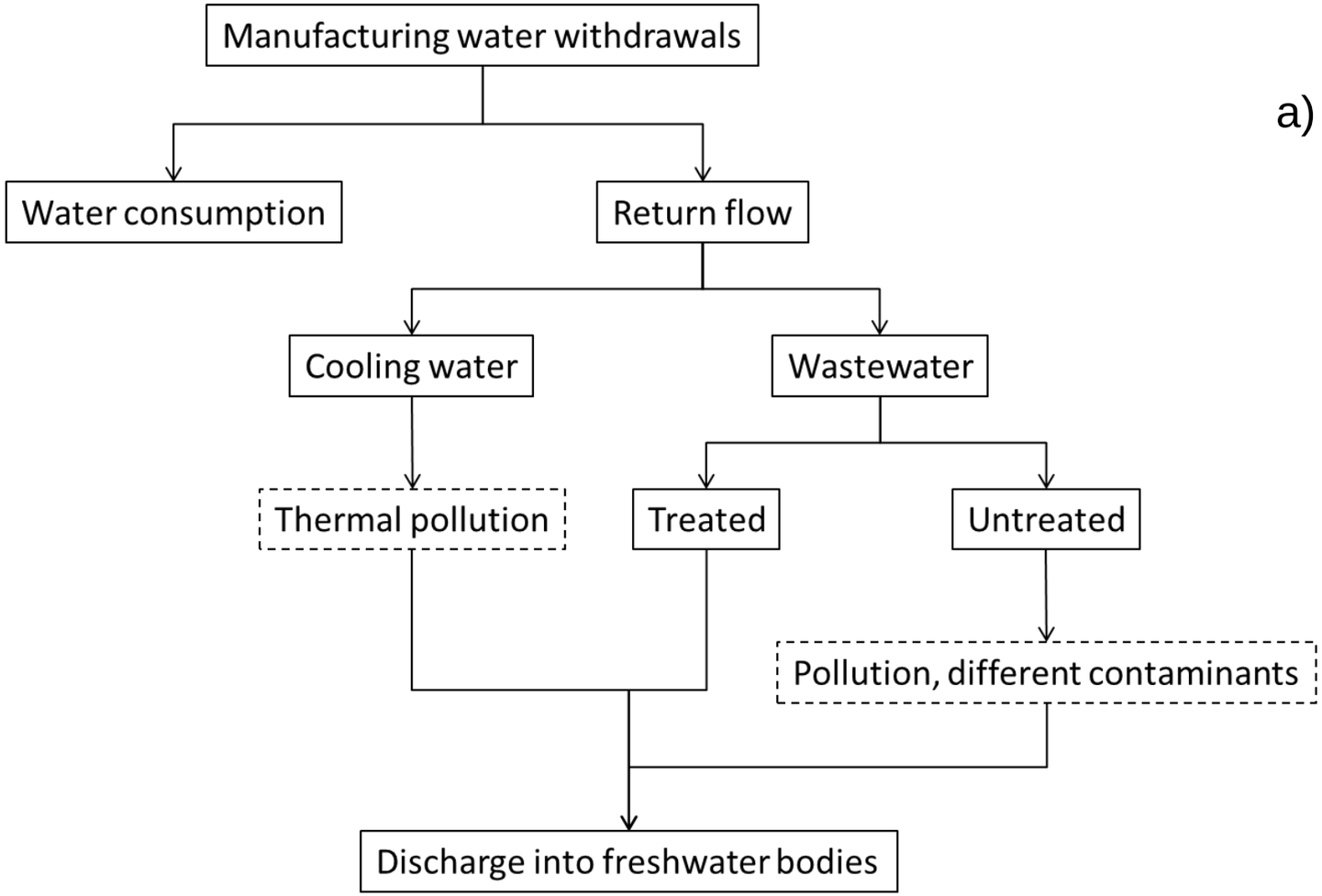
Cooling Water for Electricity Production





Flörke et al. (2013)

Manufacturing Water Use Model



a) Water withdrawals:

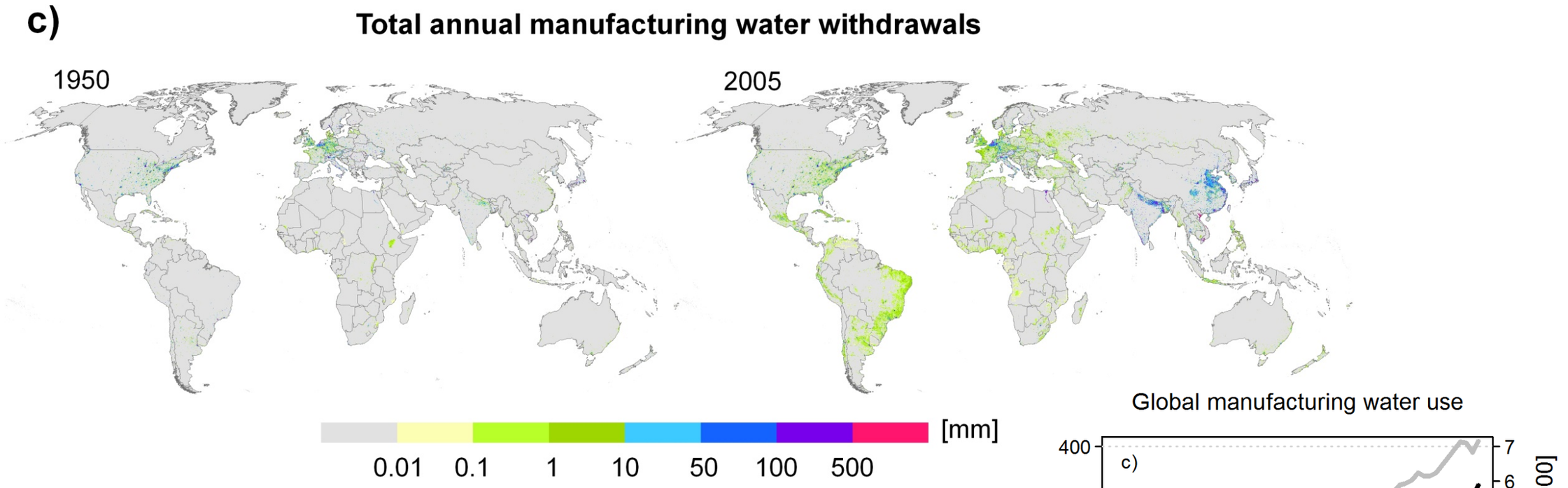
$$MWW_t = MSWI_{2005} \cdot GVA_t \cdot TC_t \qquad \left[\frac{m^3}{year}\right]$$

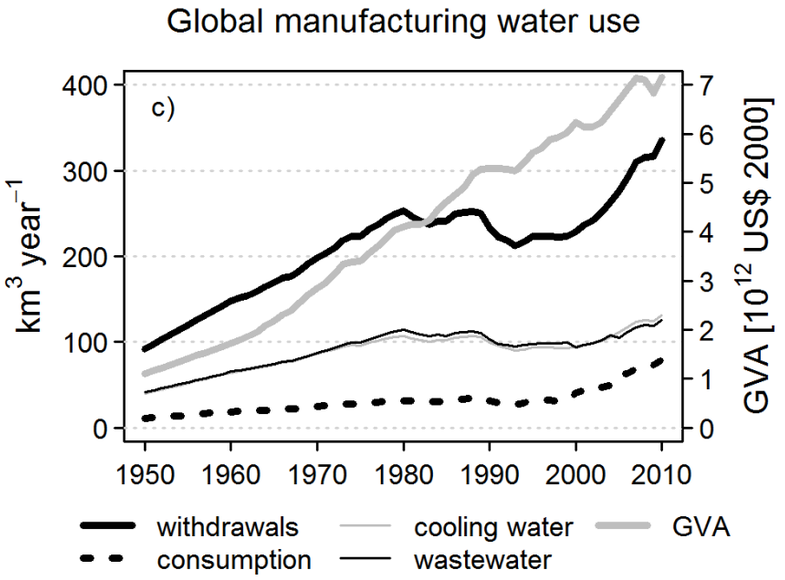
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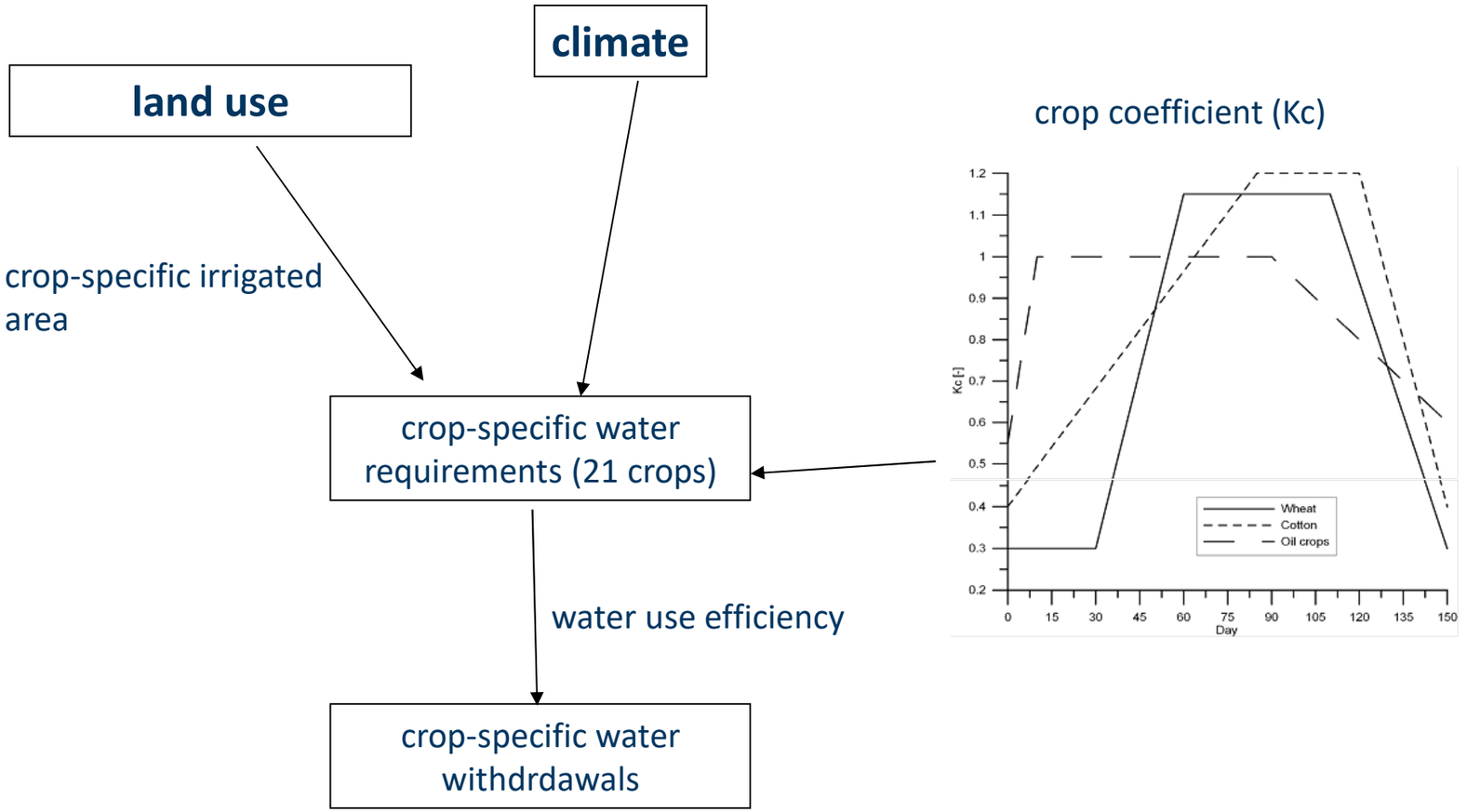
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Manufacturing Water Withdrawals

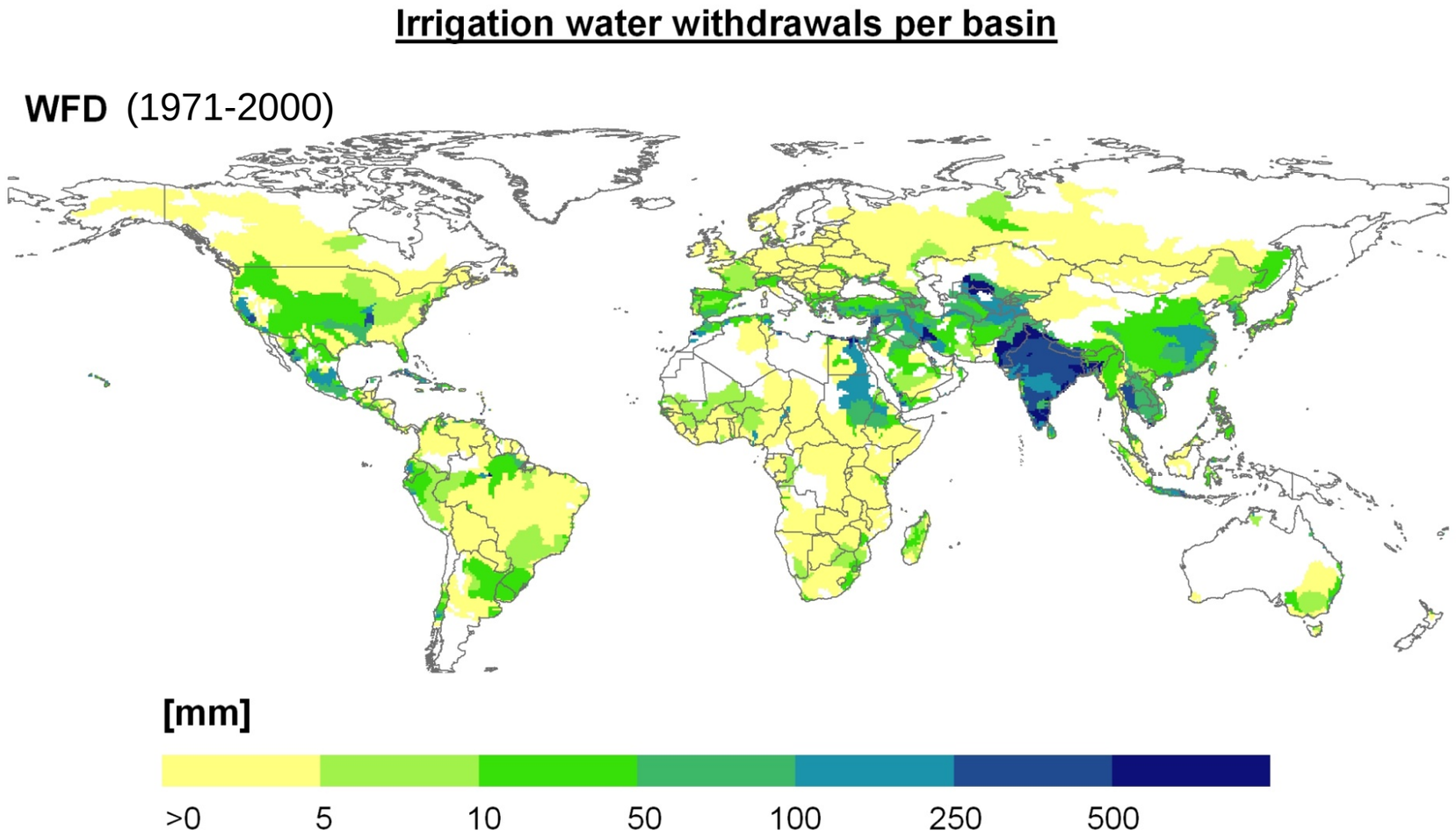




Irrigation Water Model



Irrigation Water Withdrawals



Scenario Simulations

